

Bidirectional Search

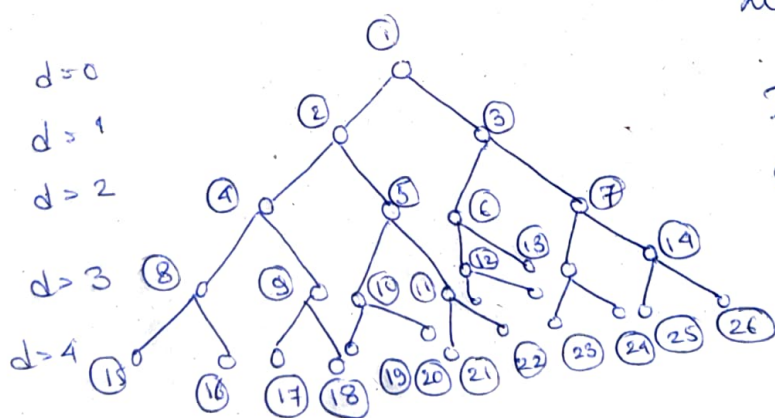
The operator of searching maps from A to B and another operator that maps from B to A.

There are other search space methods that does not follow this method.

Hence the arc starts from start state and searching for goal, one may start from goal state and by reaching the start state.

Thus if the goal exists then the two search should meet at the middle.

The forward search will maintain fringe list and so as the backward search will have. If the intersection ^{of two fringe list} is not null set then the goal is found. However the solution may not be the optimal.



Let,

Consider

Initial state = 1

Goal state = 26

Starting with simple BFS to reach to goal (26) all the nodes will be accessed.

However in bidirectional search, if BFS is used for both forward and backward search then very less no. of nodes are accessed and goal is reached.

Forward Search

Intersection

Backward Search

1

ϕ

26

1 2 3

ϕ

26 14

1 2 3 4 5 6 7

7

26 14 25 7

1 2 3 4 5 6 7

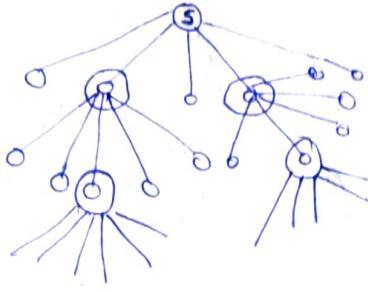


common fringe

Hence search path will be upto $d/2$ for both the search.

Beam Search

Beam
width (β)
= 2



In beam search instead of expanding the best solution (as was for Hill climbing), two best solutions are expanded.

Hence Beam width decides the no. of best solutions to be taken for expansion.

The best application of Beam Search is in speech recognition.

The main complexity in beam search is space complexity.

This is also a Heuristic search based algorithm. The local maxima problem of Hill climbing is overcome using Beam Search.

It is also an optimized version of Best First Search.